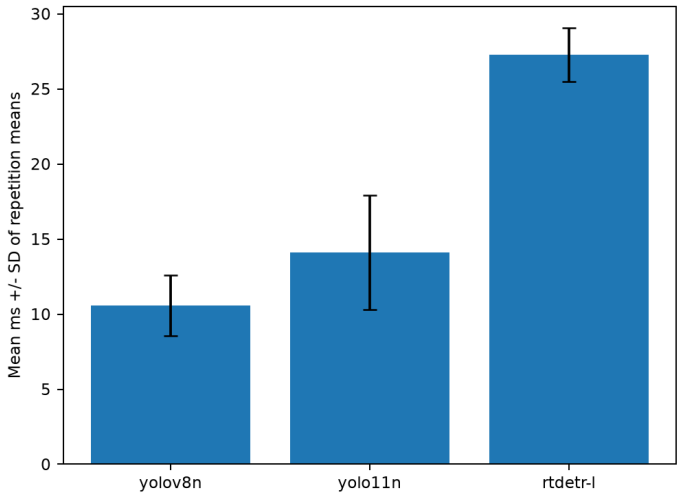
The 19 benchmark images: timing and qualitative comparison

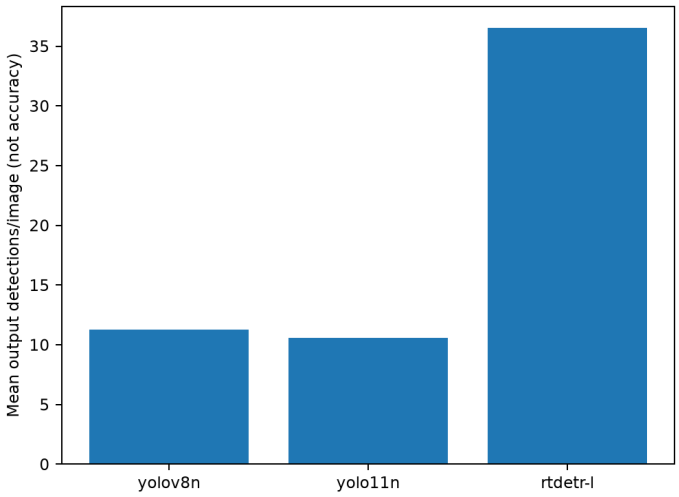
Supplement to the verified CMPE 249 Track A experiments. These exact 19 local images were used for application timing, not ground-truth accuracy evaluation. This report covers every image, with file-level timing and saved outputs for YOLOv8n, YOLO11n and RT-DETR-L.

Settings: CUDA device 0, square input 512, FP32, batch 1, confidence .25, export cap 300. YOLO uses NMS IoU .70; RT-DETR is NMS-free. YOLO letterboxes and RT-DETR stretches to the square input. All three models run in isolated processes. The timing source is the prior verified run; fresh visual prediction passes were outside timing.

Measurement: decoded images preloaded; 10 warmup calls, 5 repetitions/image, 95 timed calls/model. CUDA synchronization surrounds the full model.predict call. Disk reading, decoding, export and drawing are excluded. Framework forward time is separate. Allocated/reserved memory reports PyTorch allocator peaks. No AP, precision, recall or F1 is measured because labels are unavailable.

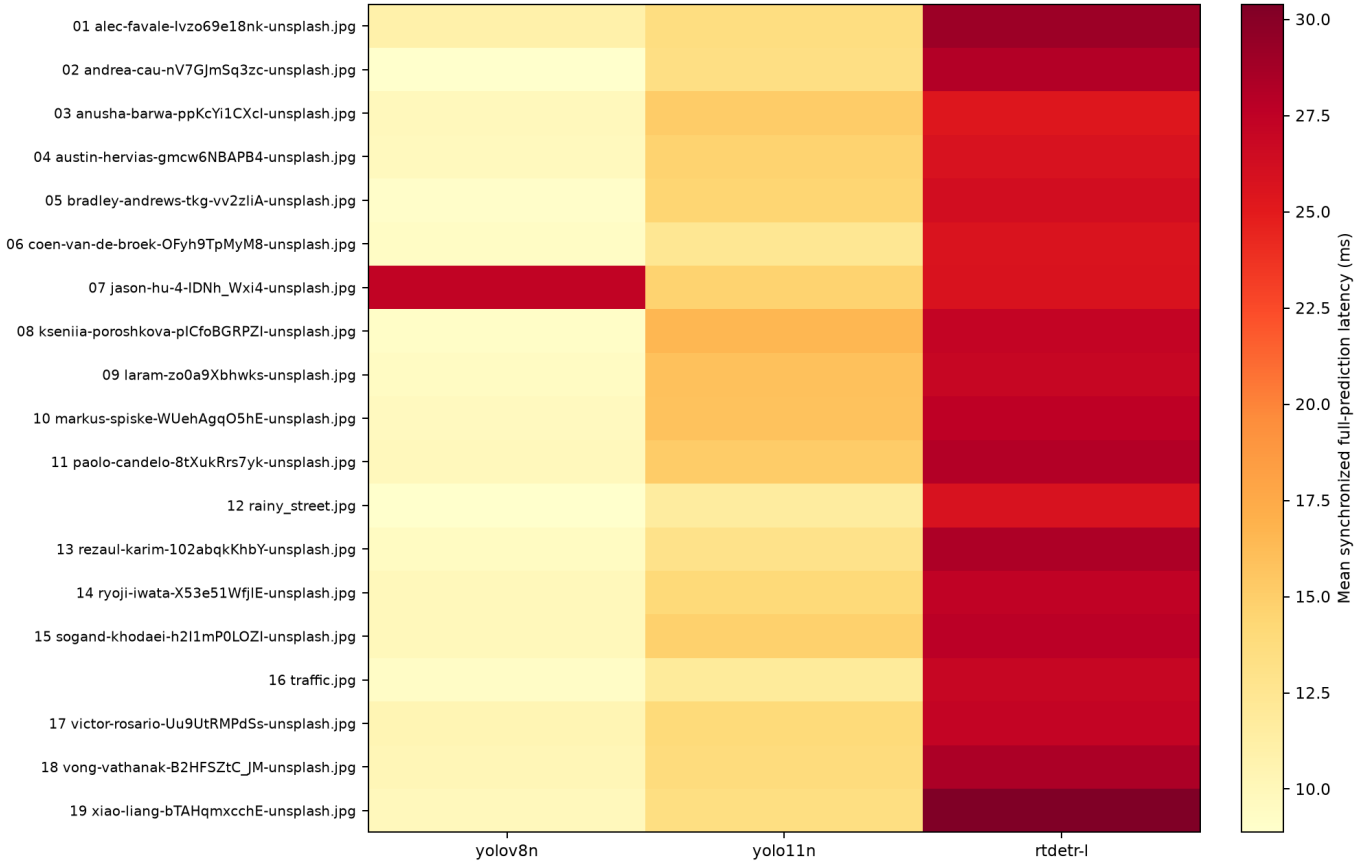
Model	Mean ms	Median	p95	FPS	Alloc / reserved MiB
yolov8n.pt	10.59	9.49	11.16	94.45	40.4 / 62.0
yolo11n.pt	14.10	12.44	23.14	70.90	68.2 / 90.0
rtdetr-l.pt	27.27	26.16	34.41	36.67	237.2 / 302.0





Per-image timing and interpretation

The heat map identifies which images had larger measured mean latency. Every cell averages five repetitions. Variation includes host preprocessing, model forward, postprocessing and system scheduling; image content alone cannot be assigned as the cause. Keep unusually slow samples in the dataset rather than selecting favorable repetitions.



RT-DETR emits more boxes at the common confidence threshold. This is not evidence of greater precision or recall. Confidence scores are not calibrated identically across models. The same object may receive different classes, box extents or scores; agreement and disagreement both need labeled adjudication.

Visual inventory: the actual images include animal pairs, cyclists, traffic, street intersections, overhead pedestrians, dense crowds, blurred subjects and a wet street. These descriptions come from inspecting the contact sheet, not from the unused image-download script. Scene-specific observations accompany every comparison page.

Inventory and image-level results

ID / filename	Dimensions	YOLOv8n ms / boxes	YOLO11n ms / boxes	RT-DETR ms / boxes
01 alec-favale-Ivzo69e18nk-unsplash.jpg	3652x5479	10.96 / 2	13.56 / 2	29.07 / 2
02 andrea-cau-nV7GJmSq3zc-unsplash.jpg	1920x1280	8.94 / 19	13.38 / 16	28.12 / 56
03 anusha-barwa-ppKcYi1CXcI-unsplash.jpg	4000x4020	9.81 / 2	15.12 / 2	25.38 / 2
04 austin-hervias-gmcw6NBAPB4-unsplash.jpg	6000x4000	9.74 / 14	14.64 / 14	25.82 / 39
05 bradley-andrews-tkg-vv2zliA-unsplash.jpg	3089x2048	9.07 / 12	14.44 / 11	26.23 / 30
06 coen-van-de-broek-OFyh9TpMyM8-unsplash.jpg	6000x4000	9.35 / 4	12.35 / 5	25.61 / 10
07 jason-hu-4-IDNh_Wxi4-unsplash.jpg	3392x4160	27.33 / 13	14.63 / 13	25.74 / 40
08 kseniia-poroshkova-pICfoBGRPZI-unsplash.jpg	4000x6000	9.17 / 5	16.61 / 8	27.25 / 23
09 laram-zo0a9Xbhwks-unsplash.jpg	4993x3328	9.39 / 13	15.87 / 8	26.99 / 55
10 markus-spiske-WUehAgqO5hE-unsplash.jpg	5760x3840	9.69 / 26	15.80 / 25	27.55 / 44
11 paolo-candelo-8tXukRrs7yk-unsplash.jpg	4253x2835	9.84 / 6	15.12 / 8	28.07 / 10
12 rainy_street.jpg	335x597	8.87 / 2	11.63 / 3	25.80 / 17
13 rezaul-karim-102abqkKhbY-unsplash.jpg	3405x5107	9.49 / 14	12.96 / 14	28.32 / 48
14 ryoji-iwata-X53e51WfjlE-unsplash.jpg	4016x6016	9.94 / 20	14.04 / 22	27.37 / 53
15 sogand-khodaei-h2I1mP0LOZI-unsplash.jpg	3376x6000	9.94 / 3	14.79 / 3	27.78 / 4
16 traffic.jpg	1740x1161	9.26 / 22	11.86 / 21	27.03 / 140
17 victor-rosario-Uu9UtRMPdSs-unsplash.jpg	3522x5283	10.35 / 13	13.97 / 10	27.22 / 32
18 vong-vathanak-B2HFSZtC_JM-unsplash.jpg	6000x4000	10.18 / 15	13.80 / 9	28.39 / 40
19 xiao-liang-bTAHqmxcchE-unsplash.jpg	9504x6336	9.84 / 9	13.43 / 7	30.40 / 49

Boxes = outputs from the untimed .25 qualitative pass; milliseconds = five-repeat mean from the original timing pass. Full distributions, model class counts and all parameter cases are in the CSV tables. Source images, manifests and input hashes preserve correspondence.

Image 01: model outputs

alec-favale-lvzo69e18nk-unsplash.jpg | 3652 x 5479 source pixels

A dog and cat lie together beside a wooden wall. Their bodies overlap. This is a useful close-range animal example; examine whether the boxes separate the two animals.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	10.96 +/- 2.20	2	cat: 1; dog: 1
yolo11n.pt	13.56 +/- 3.02	2	cat: 1; dog: 1
rtdetr-l.pt	29.07 +/- 3.68	2	cat: 1; dog: 1

Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 02: model outputs

andrea-cau-nV7GJmSq3zc-unsplash.jpg | 1920 x 1280 source pixels

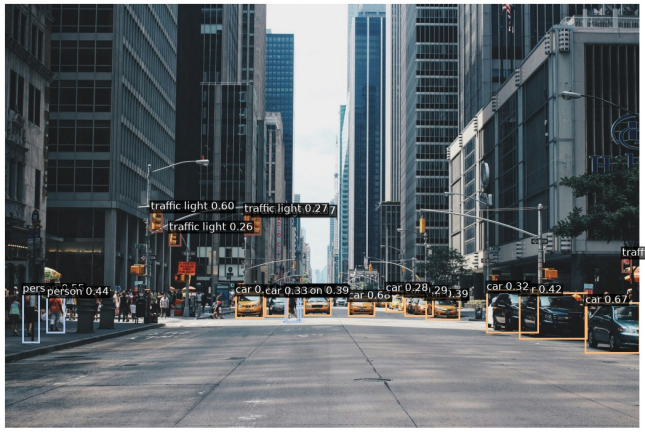
An urban intersection contains distant vehicles and pedestrians between tall buildings. Many objects occupy few pixels after the scene is resized to 512.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	8.94 +/- 0.49	19	car: 10; person: 4; traffic light: 5
yolo11n.pt	13.38 +/- 4.01	16	car: 5; person: 5; traffic light: 6
rtdetr-l.pt	28.12 +/- 3.66	56	bicycle: 1; car: 16; person: 23; traffic light: 14; truck: 1; umbrella: 1

Image 02: andrea-cau-nV7GJmSq3zc-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

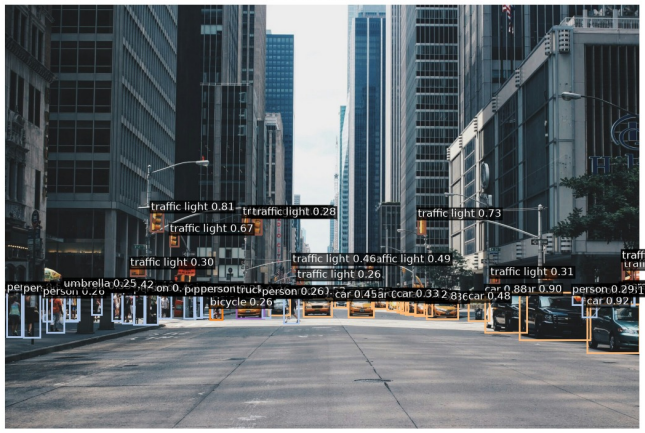
Original - no ground-truth labels supplied

yolov8n | 19 detections



yolo11n | 16 detections

rtdetr-l | 56 detections



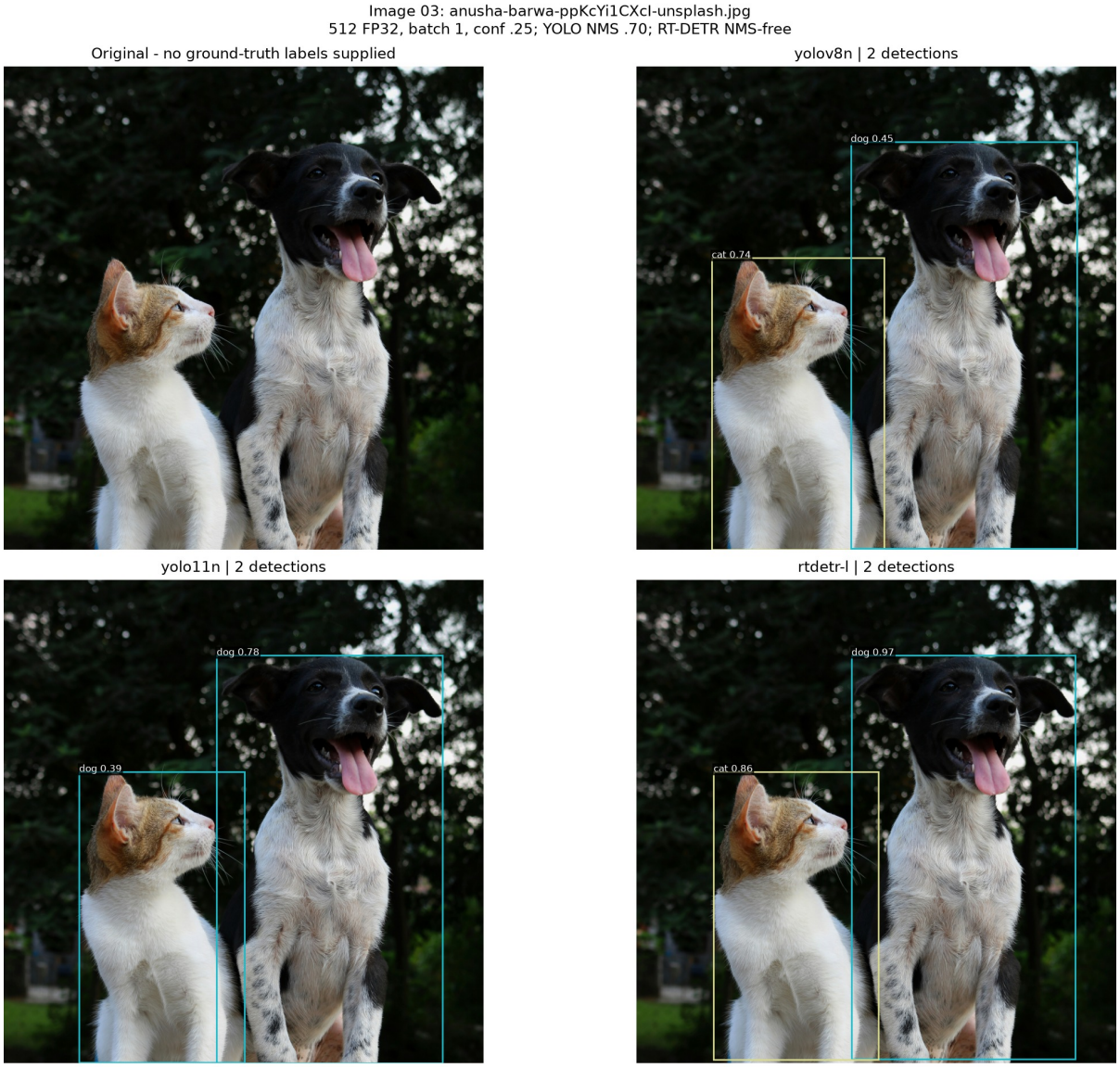
Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 03: model outputs

anusha-barwa-ppKcYi1CXcI-unsplash.jpg | 4000 x 4020 source pixels

A cat sits beside a larger dog outdoors. The animals differ in scale and parts of their outlines are close together against a dark background.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.81 +/- 0.75	2	cat: 1; dog: 1
yolo11n.pt	15.12 +/- 6.66	2	dog: 2
rtdetr-l.pt	25.38 +/- 1.24	2	cat: 1; dog: 1



Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 04: model outputs

austin-hervias-gmcw6NBAPB4-unsplash.jpg | 6000 x 4000 source pixels

A foreground pedestrian crosses an urban intersection carrying bags. Other people and vehicles are farther away, providing several object scales in one frame.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.74 +/- 0.71	14	car: 5; person: 7; traffic light: 1; truck: 1
yolo11n.pt	14.64 +/- 4.68	14	car: 4; person: 8; traffic light: 1; truck: 1
rtdetr-l.pt	25.82 +/- 0.65	39	car: 8; clock: 1; handbag: 2; person: 15; traffic light: 11; truck: 2

Image 04: austin-hervias-gmcw6NBAPB4-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

Original - no ground-truth labels supplied

yolov8n | 14 detections

yolo11n | 14 detections

rtdetr-l | 39 detections

Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 05: model outputs

bradley-andrews-tkg-vv2zliA-unsplash.jpg | 3089 x 2048 source pixels

A street view contains cars, traffic lights and small distant pedestrians. Thin poles and building details create substantial background structure.

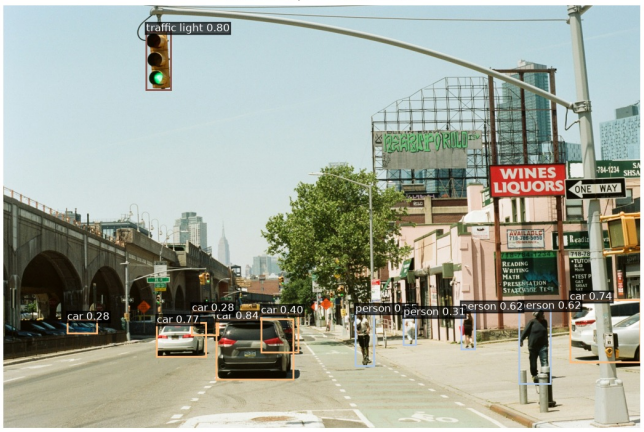
Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.07 +/- 0.57	12	car: 8; person: 3; traffic light: 1
yolo11n.pt	14.44 +/- 5.35	11	car: 6; person: 4; traffic light: 1
rtdetr-l.pt	26.23 +/- 1.10	30	bus: 2; car: 15; fire hydrant: 1; person: 6; traffic light: 5; truck: 1

Image 05: bradley-andrews-tkg-vv2zliA-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

Original - no ground-truth labels supplied

yolov8n | 12 detections

yolo11n | 11 detections



rtdetr-l | 30 detections

Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 06: model outputs

coen-van-de-broek-OFyh9TpMyM8-unsplash.jpg | 6000 x 4000 source pixels

Two cyclists ride side by side. Person and bicycle are separate COCO categories; detecting both on one rider is not itself a duplicate-object error.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.35 +/- 0.44	4	bicycle: 2; person: 2
yolo11n.pt	12.35 +/- 0.78	5	backpack: 1; bicycle: 2; person: 2
rtdetr-l.pt	25.61 +/- 1.53	10	backpack: 2; bicycle: 3; bottle: 3; person: 2

Image 06: coen-van-de-broek-OFyh9TpMyM8-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

Original - no ground-truth labels supplied

yolov8n | 4 detections

yolo11n | 5 detections

rtdetr-l | 10 detections

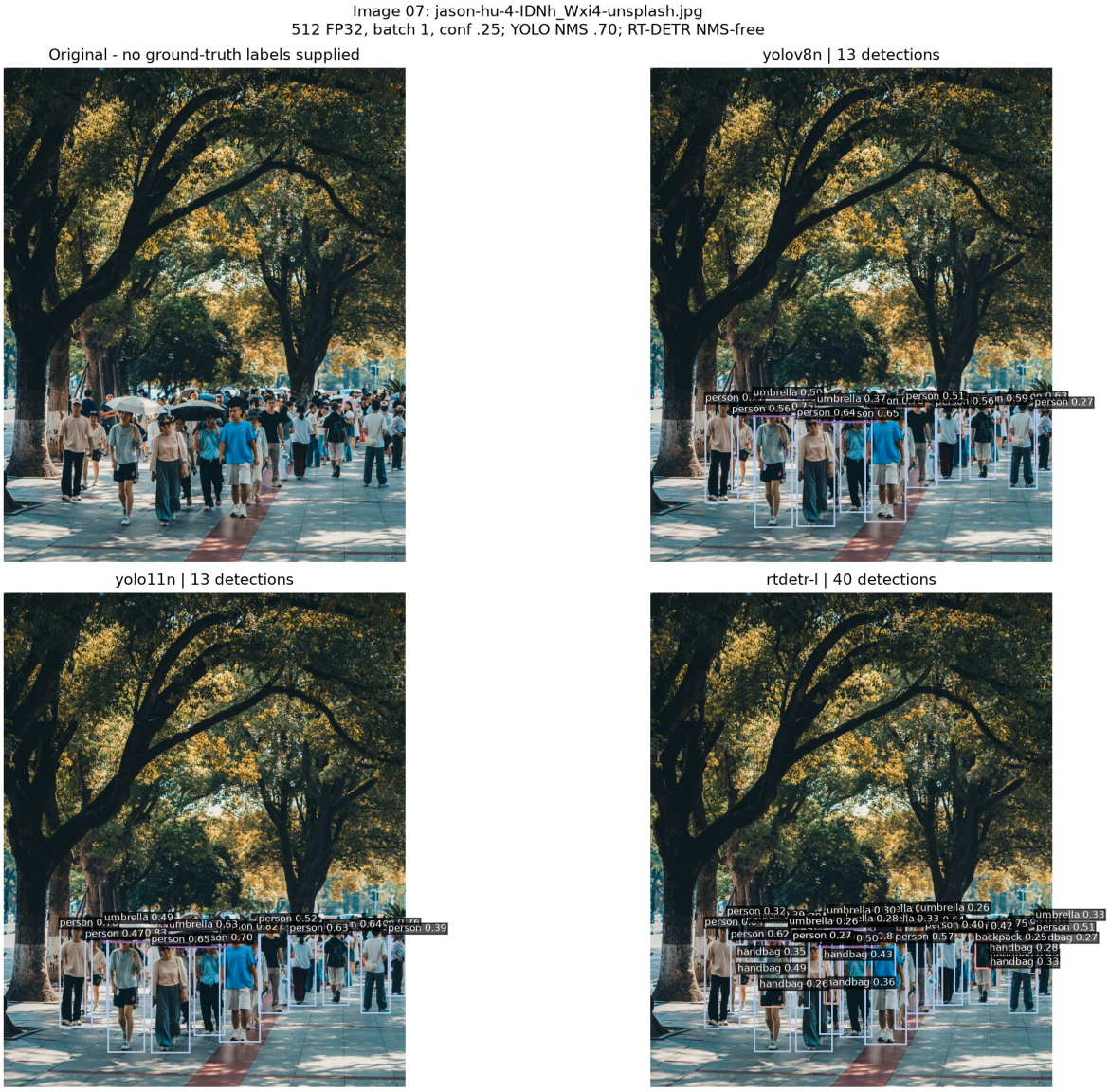
Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 07: model outputs

jason-hu-4-IDNh_Wxi4-unsplash.jpg | 3392 x 4160 source pixels

Pedestrians walk beneath a tree canopy with umbrellas and bags. People overlap and become smaller with distance. Inspected prediction panels show many more accessory boxes in RT-DETR's output.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	27.33 +/- 40.00	13	person: 11; umbrella: 2
yolo11n.pt	14.63 +/- 3.91	13	person: 11; umbrella: 2
rtdetr-l.pt	25.74 +/- 2.13	40	backpack: 1; handbag: 9; person: 19; umbrella: 11



Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 08: model outputs

kseniia-poroshkova-pICfoBGRPZI-unsplash.jpg | 4000 x 6000 source pixels

A city square includes a cyclist and small distant people. The portrait aspect ratio produces a substantially different letterboxed content region from RT-DETR's square stretching.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.17 +/- 0.31	5	bicycle: 1; car: 1; person: 3
yolo11n.pt	16.61 +/- 7.24	8	bicycle: 1; car: 1; person: 6
rtdetr-l.pt	27.25 +/- 2.39	23	bicycle: 1; car: 1; person: 20; traffic light: 1

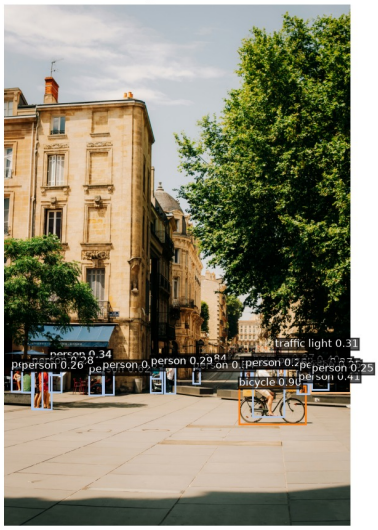
Image 08: kseniia-poroshkova-pICfoBGRPZI-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

Original - no ground-truth labels supplied

yolov8n | 5 detections

yolo11n | 8 detections

rtdetr-l | 23 detections



Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 09: model outputs

laram-zo0a9Xbhwks-unsplash.jpg | 4993 x 3328 source pixels

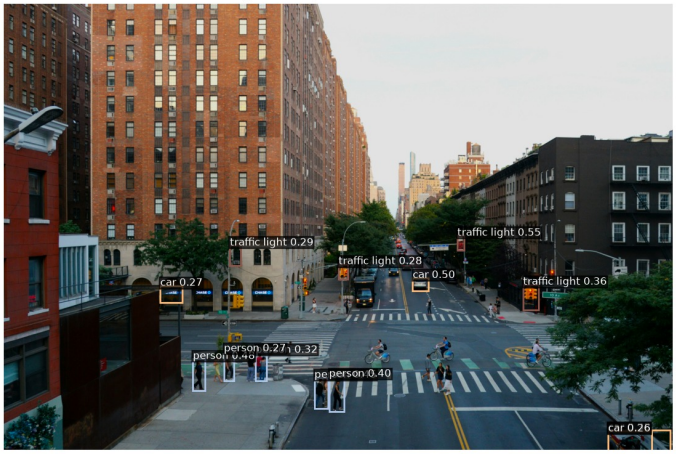
An elevated view looks down on a street intersection with pedestrians and vehicles. Many candidates are small relative to the buildings and road surface.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.39 +/- 0.18	13	car: 4; person: 5; traffic light: 4
yolo11n.pt	15.87 +/- 5.50	8	car: 1; person: 7
rtdetr-l.pt	26.99 +/- 3.88	55	backpack: 1; bicycle: 4; bus: 1; car: 10; motorcycle: 1; person: 22; traffic light: 14; truck: 2

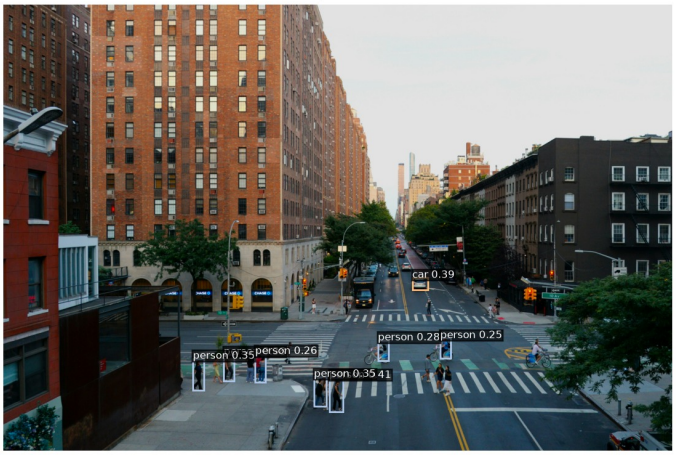
Image 09: laram-zo0a9Xbhwks-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

Original - no ground-truth labels supplied

yolov8n | 13 detections



yolo11n | 8 detections



rtdetr-l | 55 detections

Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 10: model outputs

markus-spiske-WUehAgqO5hE-unsplash.jpg | 5760 x 3840 source pixels

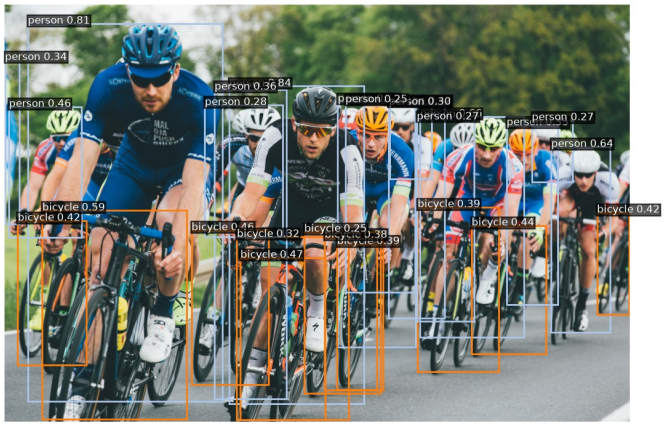
A closely grouped line of cyclists overlaps in depth. Bicycles, riders, wheels and helmets create repeated structures; inspect localization and class assignments separately.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.69 +/- 0.59	26	bicycle: 11; person: 15
yolo11n.pt	15.80 +/- 5.23	25	bicycle: 10; person: 15
rtdetr-l.pt	27.55 +/- 3.83	44	bicycle: 23; person: 21

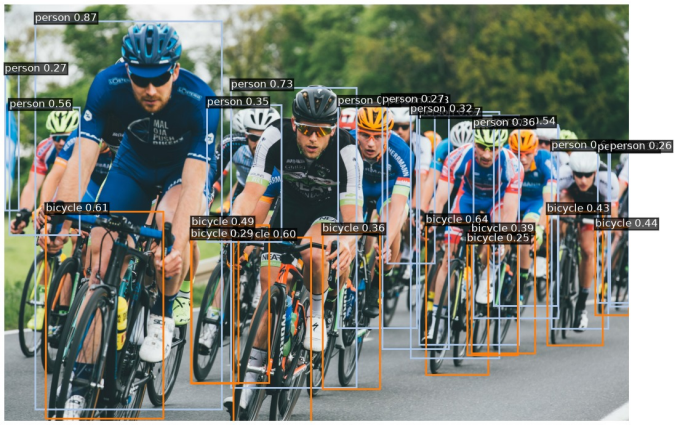
Image 10: markus-spiske-WUehAgqO5hE-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

Original - no ground-truth labels supplied

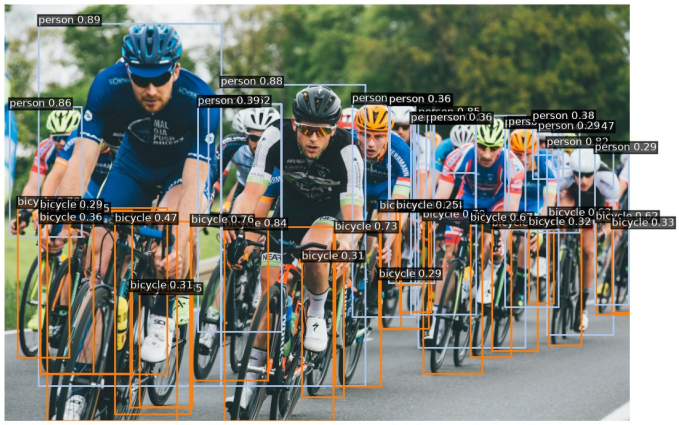
yolov8n | 26 detections



yolo11n | 25 detections



rtdetr-l | 44 detections



Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 11: model outputs

paolo-candelo-8tXukRrs7yk-unsplash.jpg | 4253 x 2835 source pixels

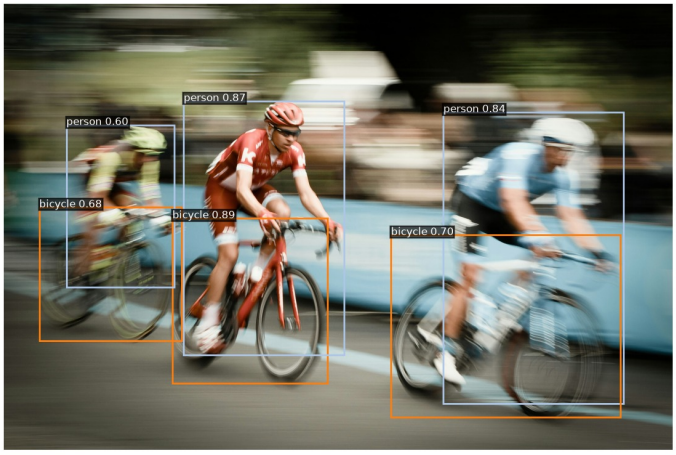
Cyclists are shown with pronounced background and motion blur. The foreground rider is more distinct than some surrounding riders. Output disagreement here is a qualitative observation, not a measured blur-robustness score.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.84 +/- 0.95	6	bicycle: 3; person: 3
yolo11n.pt	15.12 +/- 3.91	8	bicycle: 3; person: 5
rtdetr-l.pt	28.07 +/- 3.42	10	backpack: 2; bicycle: 4; person: 4

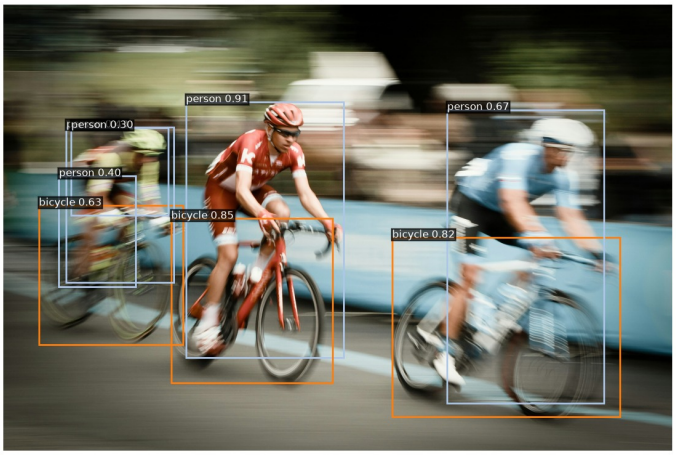
Image 11: paolo-candelo-8tXukRrs7yk-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

Original - no ground-truth labels supplied

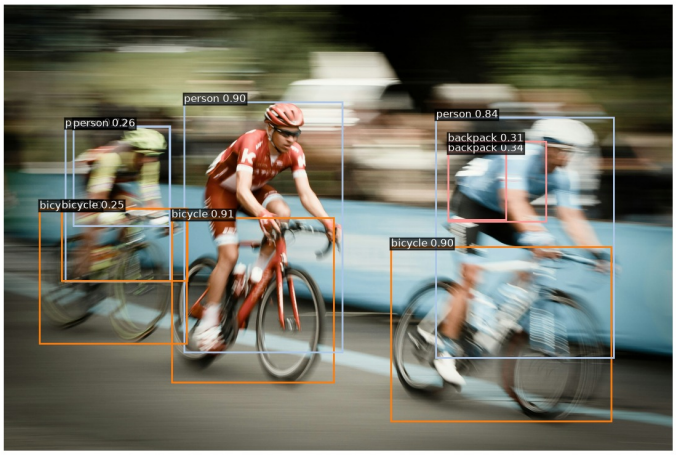
yolov8n | 6 detections



yolo11n | 8 detections



rtdetr-l | 10 detections



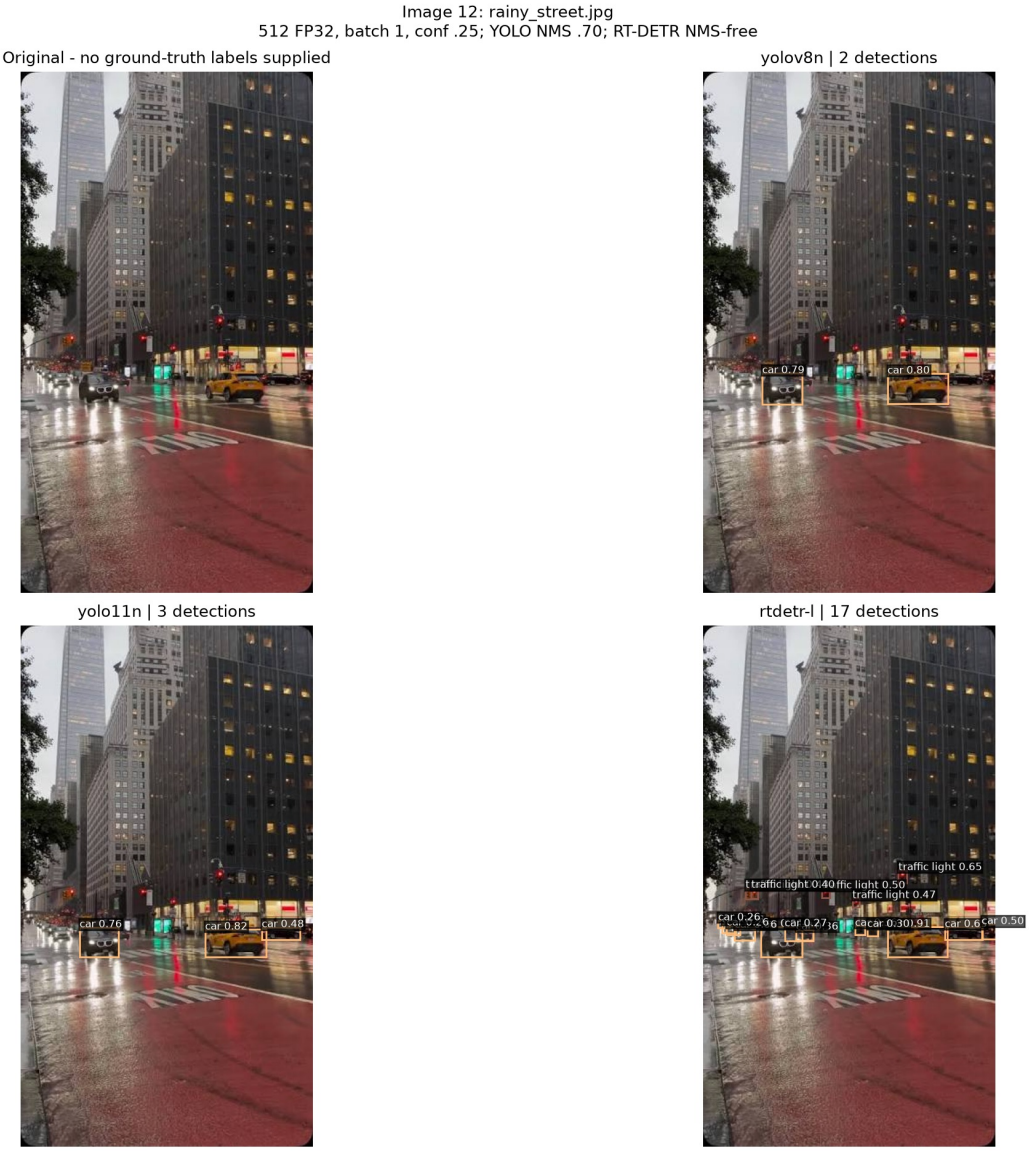
Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 12: model outputs

rainy_street.jpg | 335 x 597 source pixels

A wet urban street has reflective road markings and vehicles under relatively dim illumination. Reflections and bright vehicle lights alter contrast.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	8.87 +/- 0.55	2	car: 2
yolo11n.pt	11.63 +/- 2.36	3	car: 3
rtdetr-l.pt	25.80 +/- 2.15	17	car: 12; traffic light: 5



Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 13: model outputs

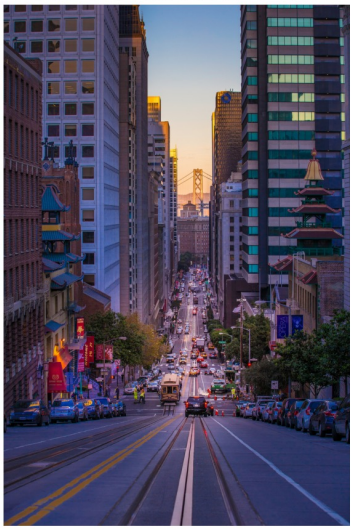
rezaul-karim-102abqkKhbY-unsplash.jpg | 3405 x 5107 source pixels

A long urban street is viewed between tall buildings, with warm light at the horizon and small vehicles in the distance. Strong perspective produces a large scale range.

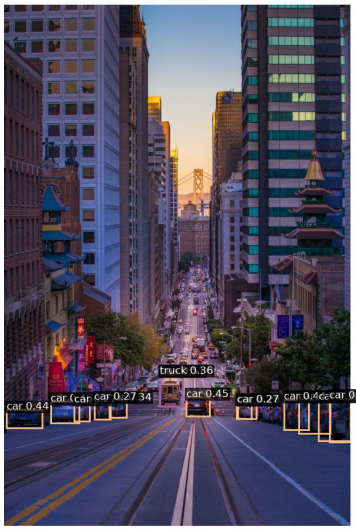
Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.49 +/- 0.39	14	bus: 1; car: 12; truck: 1
yolo11n.pt	12.96 +/- 4.22	14	car: 14
rtdetr-l.pt	28.32 +/- 3.61	48	bus: 4; car: 30; fire hydrant: 1; person: 5; traffic light: 3; truck: 5

Image 13: rezaul-karim-102abqkKhbY-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

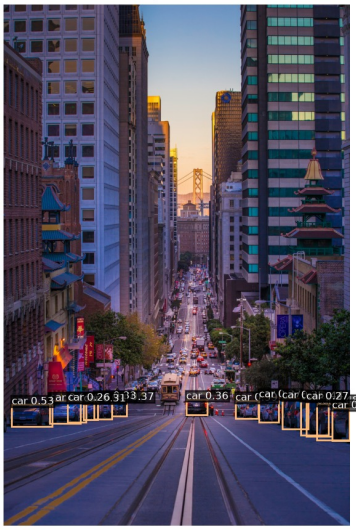
Original - no ground-truth labels supplied



yolov8n | 14 detections



yolo11n | 14 detections



rtdetr-l | 48 detections

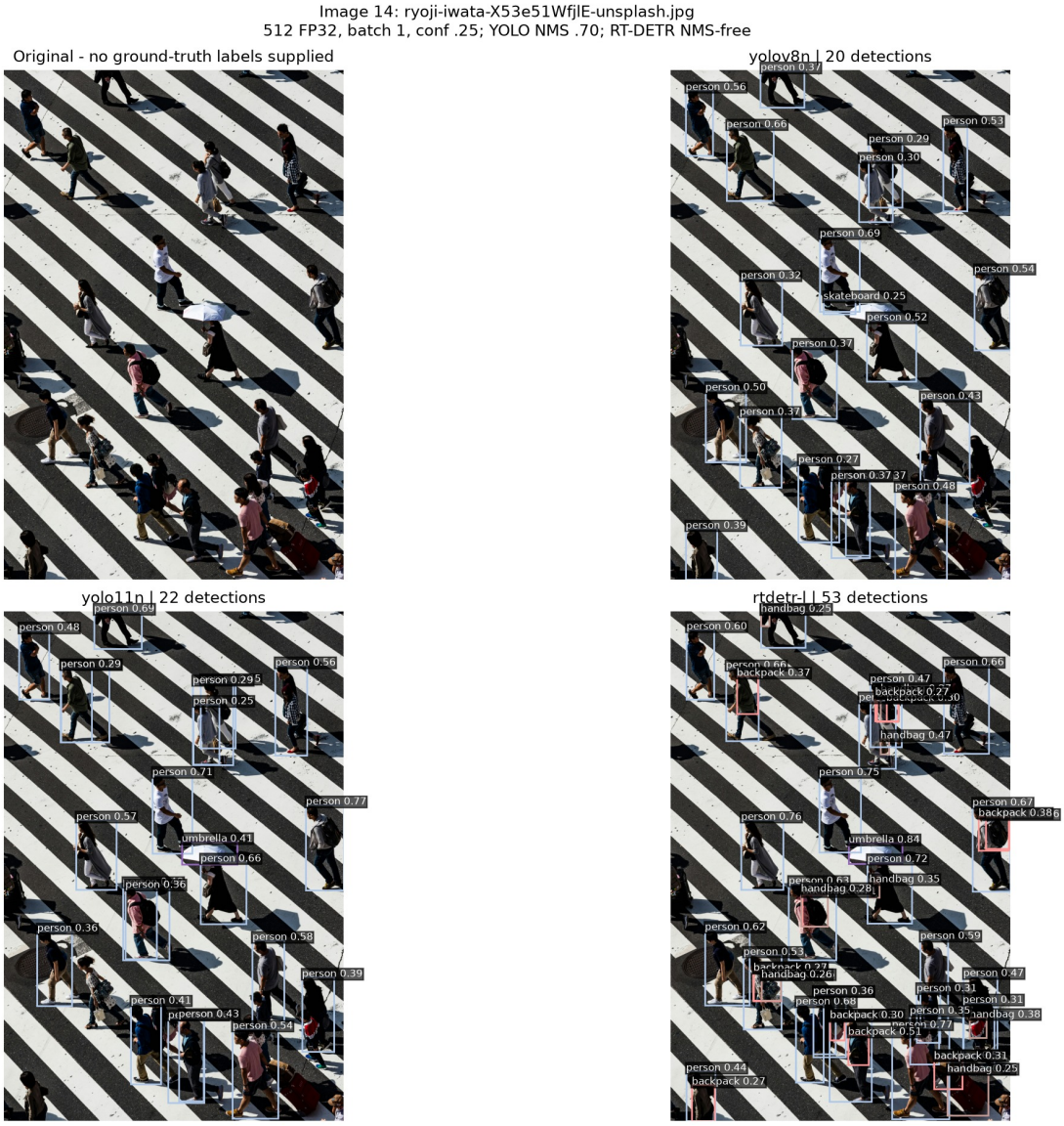
Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 14: model outputs

ryoji-iwata-X53e51WfjlE-unsplash.jpg | 4016 x 6016 source pixels

People cross broad zebra stripes from an elevated viewpoint. The repeated stripe pattern and top-down body orientations differ from a typical eye-level portrait.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.94 +/- 0.41	20	person: 19; skateboard: 1
yolo11n.pt	14.04 +/- 3.57	22	person: 21; umbrella: 1
rtdetr-l.pt	27.37 +/- 4.49	53	backpack: 14; handbag: 13; person: 24; suitcase: 1; umbrella: 1



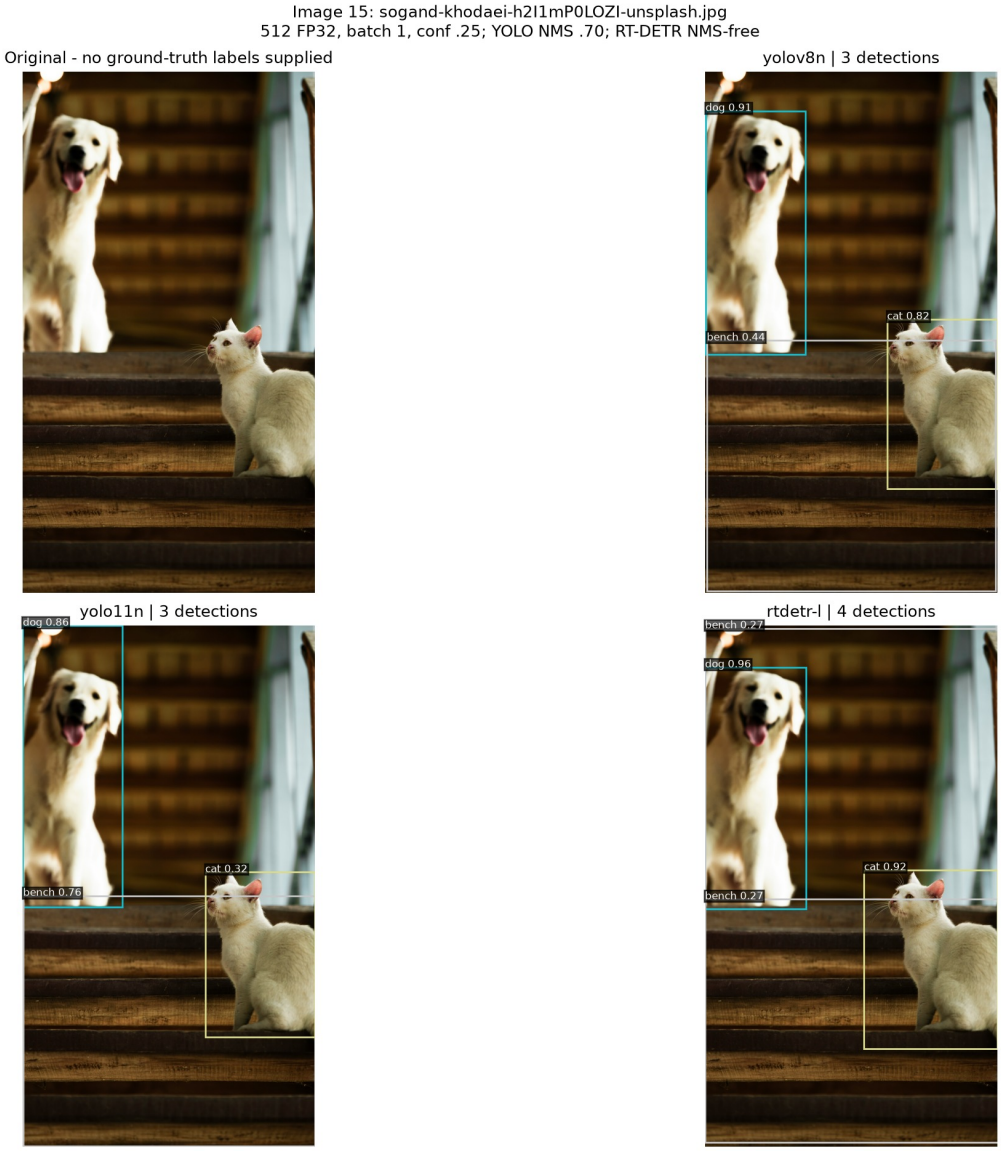
Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 15: model outputs

sogand-khodaei-h2I1mP0LOZI-unsplash.jpg | 3376 x 6000 source pixels

A dog and cat occupy different steps on a staircase. Uneven lighting and the stair edges provide a second animal scene distinct from the close overlapping examples.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.94 +/- 1.13	3	bench: 1; cat: 1; dog: 1
yolo11n.pt	14.79 +/- 5.17	3	bench: 1; cat: 1; dog: 1
rtdetr-l.pt	27.78 +/- 4.21	4	bench: 2; cat: 1; dog: 1



Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 16: model outputs

traffic.jpg | 1740 x 1161 source pixels

Dense traffic contains cars, buses and two-wheelers at several depths. Inspected RT-DETR outputs are substantially denser and include many low-confidence overlapping labels. The count difference does not establish correctness.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.26 +/- 0.97	22	bus: 3; car: 14; motorcycle: 3; person: 1; truck: 1
yolo11n.pt	11.86 +/- 1.26	21	bus: 2; car: 15; motorcycle: 2; person: 1; truck: 1
rtdetr-l.pt	27.03 +/- 4.64	140	bus: 7; car: 30; motorcycle: 38; person: 37; traffic light: 2; truck: 26

Image 16: traffic.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

Original - no ground-truth labels supplied

yolov8n | 22 detections

yolo11n | 21 detections

rtdetr-l | 140 detections

Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 17: model outputs

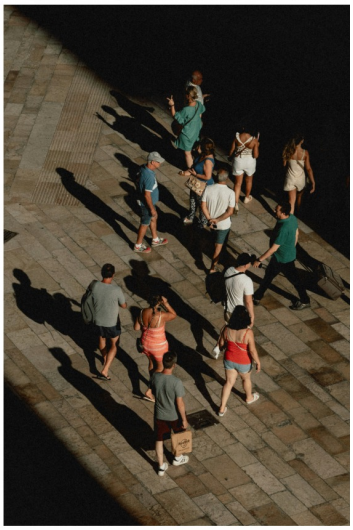
victor-rosario-Uu9UtRMPdSs-unsplash.jpg | 3522 x 5283 source pixels

People are viewed from above with strong directional lighting and long shadows. Shadows and partial overlap change the visible person silhouettes.

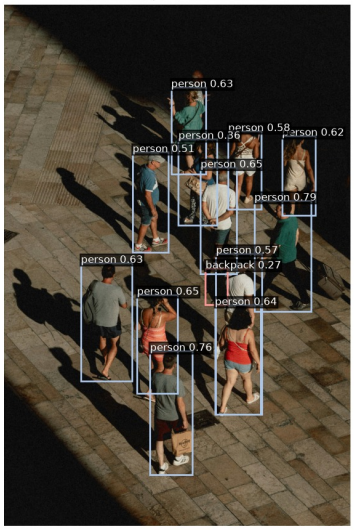
Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	10.35 +/- 0.94	13	backpack: 1; person: 12
yolo11n.pt	13.97 +/- 5.61	10	person: 10
rtdetr-l.pt	27.22 +/- 5.36	32	backpack: 4; handbag: 13; person: 15

Image 17: victor-rosario-Uu9UtRMPdSs-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

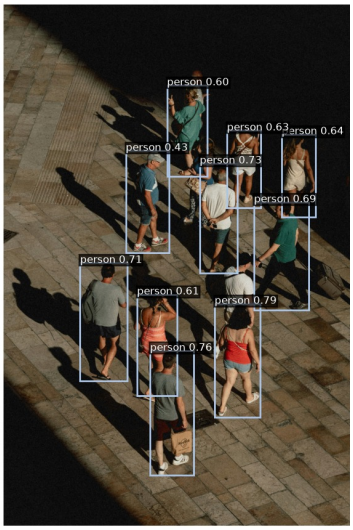
Original - no ground-truth labels supplied



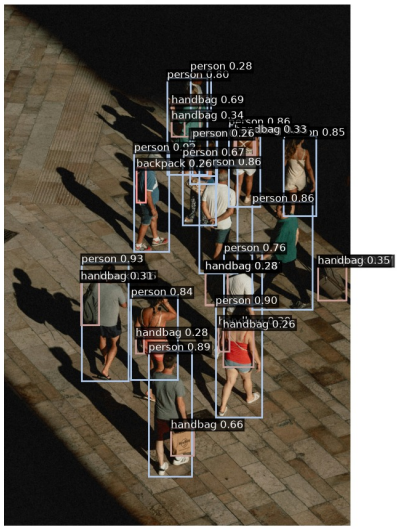
yolov8n | 13 detections



yolo11n | 10 detections



rtdetr-l | 32 detections



Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 18: model outputs

vong-vathanak-B2HFSZtC_JM-unsplash.jpg | 6000 x 4000 source pixels

A pedestrian crowd contains visibly blurred people and a more distinct foreground figure. Blur and overlap coexist, so this image cannot isolate either factor experimentally.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	10.18 +/- 0.85	15	handbag: 1; person: 14
yolo11n.pt	13.80 +/- 5.19	9	person: 9
rtdetr-l.pt	28.39 +/- 2.92	40	handbag: 11; person: 26; suitcase: 1; traffic light: 2

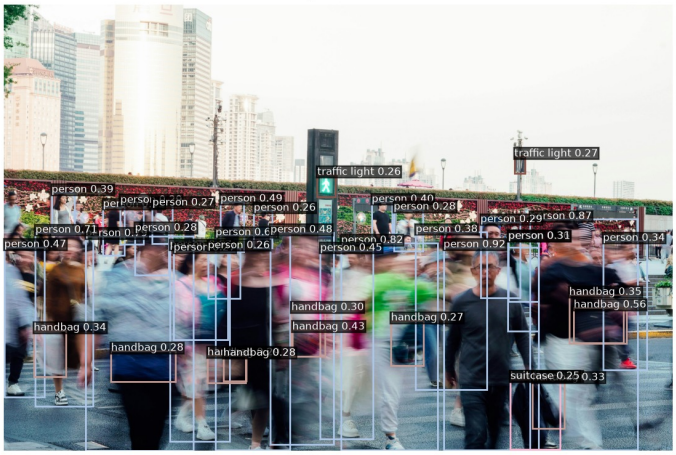
Image 18: vong-vathanak-B2HFSZtC_JM-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

Original - no ground-truth labels supplied

yolov8n | 15 detections

yolo11n | 9 detections

rtdetr-l | 40 detections



Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.

Image 19: model outputs

xiao-liang-bTAHqmxcchE-unsplash.jpg | 9504 x 6336 source pixels

A very dense crossing contains many distant pedestrians. Inspected outputs concentrate on a subset of larger figures; no full annotation was made, so no recall or missed-person count is reported.

Detector	Mean +/- SD ms	Output boxes	Classes and counts
yolov8n.pt	9.84 +/- 0.76	9	person: 9
yolo11n.pt	13.43 +/- 3.43	7	person: 7
rtdetr-l.pt	30.40 +/- 2.97	49	car: 1; handbag: 6; person: 38; traffic light: 4

Image 19: xiao-liang-bTAHqmxcchE-unsplash.jpg
512 FP32, batch 1, conf .25; YOLO NMS .70; RT-DETR NMS-free

Original - no ground-truth labels supplied

yolov8n | 9 detections

yolo11n | 7 detections

rtdetr-l | 49 detections

Original: upper left. Other panels: class labels and scores. Full-size files are in figures/. No ground-truth accuracy or error count is inferred from these visual outputs.